Amnesty and Land Grabbing in the Amazon*

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Abstract

Write abstract.

Keywords:

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1 Introduction

Brazil's 2012 forest code provided amnesty to illegal deforestation in the Amazon for properties of less than...if deforestation occurred before 2009.

It was a contentious new law supported by *ruralistas* (agribusiness) and opposed by a coalition of environmental groups and scientists.

"Land regularization restores citizenship to farmers, combats land grabbing, drastically reduces land conflicts and violence, and contributes decisively to reducing deforestation" President Lula (2010)

"The policy may solve a practical, short-term problem, but it creates a negative incentive. The message is that land invasion and deforestation will always be forgiven." Paulo Barreto (Imazon, 2009)

Supporters argued that the new legislation would incorporate previously illegal properties into a legal framework, promote greater compliance over time, and encourage environmental restoration where feasible (cite). Critics, however, contended that granting amnesty for past illegal deforestation could set a dangerous precedent, potentially encouraging future land grabbing and unlawful clearing of forests (cite).

In this paper we examine empirically the ex-post validity of these opposing claims. If expectations were not affected, we could simply compare differences in the outcome around the law change-date between properties eligible for forgiveness to those that were ineligible. The possibility of a change in expectations requires a more careful treatment of ineligible areas, since land-grabbers may expect that some ineligible areas may soon benefit from another amnesty.

Pedro: Estamos usando estas 4 classes ou merging (1) (2) e (3)? De todo modo é melhor inicialmente considerar as 4 classes Addressed!

In many settings, Students not qualifying for debt forgiveness may borrow more in anticipation of future change in eligibility

2 Setting and Data

This section describes the main policy setting we explore — a series of amnesties granted to land-grabbers illegally occupying public land in the Amazon — as well as our main data sources and variable construction. We combine data from dozens of sources in the paper, from public records to satellite images. More details on those are available in the Appendix B.

2.1 Land Use in the Amazon

The Brazilian Amazon covers an area of 420 million hectares, approximately half the size of the continental US. Almost all forest area is under federal or state jurisdiction, with land broadly divided into two groups: areas under special protection (e.g., indigenous reservations, national parks, biological reserves) and undesignated lands. Undesignated areas are public forests waiting for an official use destination, comprising a massive area of almost 60 million hectares and storing the equivalent of one year of total global carbon emissions. Occupation of undesignated areas is illegal and, as a consequence, any deforestation in those areas is also illegal.

The vast amounts of Amazon forest without official designation have long been a concern for the Brazilian government and environmentalists. Land grab in those areas is rampant, and deforestation rates tend to be much higher than in the rest of the Amazon. Based on the satellite data we introduce below and conversations with environmental watchdogs, we estimate well over a hundred thousand individuals illegally occupying land in undesignated areas with evidence of ongoing or recent deforestation. Although some of these squatters have occupied small plots of land to produce agricultural produce, the majority of deforested and not abandoned land in undesignated areas is used for cattle farming, with many individuals claiming ownership over more than one parcel.

To address the complex land use situation in the Amazon, the Brazilian government has combined a menu of policies including direct repression and more accommodating strategies. On the repressive side, the country has invested in real-time monitoring systems and increased the scope and weight of sanctions against illegal deforestation and other environmental violations. However, enforcement effectiveness is heavily based on obligations tied to property rights of rural properties, such that illegally occupied areas

make enforcement more difficult.[1] Despite a robust monitoring system, enforcement is costly and its intensity varies strongly depending on different administrations' degrees of commitment to preservation (e.g., **?** and **?**.). In practice, expropriation of squatters is rare and it is very difficult to prevent their return or dislocation to clear forest in another public area.

In recognition of these challenges, policymakers started to acknowledge the possibility of "accommodating" certain individuals occupying public land, granting them legal ownership of their rural parcels with the ultimate goal of environmental preservation. With legal ownership, rural land owners must comply with strict land use and deforestation thresholds and environmental sanctions are more easily implemented. In this context, the Brazilian government implemented an amnesty for certain land-grabbers in 2009. In 2017, a new amnesty was introduced with even more lenient qualifying criteria.

2.2 Land-Grabbing Amnesty in the Amazon

2.2.1 The 2009 Land-Grabbing Amnesty

Context— In 2009, a reform named *Programa Terra Legal* offered amnesty to a certain group of land-grabbers in federal undesignated land. The goal of the program was to target "long-established" squatters, allowing only parcels smaller than 1,500 hectares and that had been occupied at least since 2004 to be eligible for amnesty. The policy size threshold was much larger than most regional definitions of a small property used by the Brazilian government (anywhere from 20 to 200 hectares). For comparison, the average farm size in the US is under 200 hectares. Moreover, most land titles were given for free or at virtually no cost to squatters. Ownership titles also imposed no or very loosely defined constraints on future sales. Finally, no additional environmental rules or targets were tied to the titles other than the standard requirements applicable to any landowner in the Amazon.

The amnesty was first introduced through Provisional Measure 458, signed by President Lula in February 2009, and was enacted into law by Congress in June 2009 (Law 11952). The Land-Grabbing Amnesty was contentious, supported by the Lula Administration and *ruralistas* (agribusiness), while opposed by a coalition of environmental groups and scientists. The following contemporaneous quotes provide a summary of the opposing viewpoints between the two groups:

[1] For instance, Brazil's advanced deforestation warnings system is more effective if affected areas can be linked to specific individuals, and there are strict deforestation ceilings and other land-use rules that legal rural properties need to comply to maintain their land title.

"*Land regularization restores citizenship to farmers, combats land grabbing, drastically reduces land conflicts and violence, and contributes decisively to reducing deforestation.*" — President Lula (2010)

"*The policy may solve a practical, short-term problem, but it creates a negative incentive. The message is that land invasion and deforestation will always be forgiven.*" — Paulo Barreto (Imazon, 2009)

Clearly, advocates for the Amnesty focused on the cost and limited success of previous policies that attempted to resolve land conflict in the Amazon and curb illegal deforestation. In opposition, environmentalists were concerned that the policy induced moral hazard, potentially leading to more land-grabbing and deforestation under the expectation of another amnesty in the future. The potential intertemporal effect of this policy is a key feature we explore in the paper.

Policy roll-out and implementation— The program implementation was slow and riddled with technical and logistical problems. Despite a large number of requests for amnesty eligibility (see Figure A.1), only a small number of titles were given in the program's first year and claims took on average 3 years to be processed. Several of these setbacks involved an initial step in which the property boundaries submitted by applicants had to be verified and audited, as well as the need to establish occupation history and its compliance with the eligibility criteria. Many requests from different squatters included proposed property lines that overlapped, further requiring spatial conflict resolution by the government officials working on the program's roll-out.

Because program participation is endogenous, which land-grabbers are ultimately granted amnesty matter for the direct effects of the policy. This is determined by which eligible squatters decide to apply and receive forgiveness. Squatters with lower deforestation propensity may be more likely to apply, as those whose occupied areas have already been deforested more aggressively may see compliance costs as too high. This would select land-grabbers who would already deforest less absent the Amnesty, potentially reducing the positive effects expected by the Brazilian government. Similarly, property rights can raise the incentives to extract surplus through costly investment (i.e., deforesting) as uncertainty over who captures gains from land use decreases. This suggests the possibility that treatment effects from Amnesty may even increase deforestation if high-uncertainty squatters are more likely to apply.

To shed some light into Amnesty take-up, Table 2 shows application and titling rates for land-grabbers eligible to apply for the 2009 Amnesty. Application rates have a lower bound of approximately 10%, while until the program was into effect without any modifications (2016), 42% of squatters with property boundaries audited were given ownership titles.[2] Columns (1) and (2) analyze how various property characteristics predict take-up. While squatters occupying the land longer did not seem to be more likely to apply for amnesty, those in smaller areas were far more likely to apply. Since the program had as a goal to curb deforestation through ownership, we also check whether the share of deforested area as of 2008 predicts applications.

We find a concave pattern between pre-existing deforested rate and amnesty applications. Coefficient estimates suggest that higher deforestation rates predict a higher probability of amnesty requests until a deforested share of $\widehat{0.047}/[-2(\widehat{-0.052})] \approx 45\%$, after which more deforestation is associated with lower application rates. Because these estimates are conditional on history of occupation, they are consistent with the following interpretation. For land-grabbers who invaded forest areas in the same year, those with larger investments may see property rights as a way to guarantee private returns on deforestation, making them more likely to apply for amnesty. But those who deforested at higher rates in the same period appear to weigh the costs from legalization more heavily, becoming less likely to apply.

Columns (3) and (4) in Table 2 show conditional amnesty granted to applicants (which amounts to about 42% of requests that were audited by the government). Overall, areas with a longer history of occupation (going back to 1989) are less likely to receive amnesty, with a nonlinear response where those with more recent occupation become more likely to be titled. There is also some evidence that amnesty rates for properties smaller than 100 hectares had higher success rates. We see these descriptive patterns as more likely to be picking-up inefficiencies during the roll-out process than a preference by policymakers for those attributes.

2.2.2 The 2017 Land-Grabbing Amnesty

Context— In 2017, a new Amnesty was passed into law (converted from Provisional Measure 759) by the Temer Administration. Law 13465, signed in July, expanded amnesty eligibility for land-grabbers in the Amazon: invasions until 2011 and with areas up to 2,500 hectares were allowed to qualify for

[2] We can only match Amnesty requests to our eligible property data based on geography entries of properties that had their limits surveyed by the government.

amnesty under the same terms of the existing *Programa Terra Legal* from 2009. The property threshold size of 2,500 was seen as excessively large by critics — approximately only 1% of registered farms in Brazil and US, for reference — which influenced the law's informal name of "Land-Grabbing Act". The updated occupation criteria also made land-grabbing occurring after the first Amnesty, and potentially a response to it, eligible for forgiveness in the new law.

Implementation and aftermath— Certain aspects of the 2017 Land-Grabbing Amnesty were challenged in court, which nonetheless remained valid. As early as 2019 a new, third wave of amnesty was proposed by the Bolsonaro Administration, eliminating occupation tenure and area requirements altogether. This would effectively grant the possibility of amnesty to any land-grabber in undesignated public areas in the Amazon. Provisional Measure 910 expired without being enacted into law by Congress, leaving the 2017 eligibility cutoffs into effect.

2.3 Measuring Deforestation in the Amazon

Tracking deforestation in the Amazon is crucial to study the consequences of land-grabbing amnesties and to determine the patterns of human occupation in an area. We use high-resolution maps of land cover from *MapBiomas* developed based on Landsat 30×30-meter images. The Amazon biome is covered by several billion pixels of this scale, each being assigned a specific color code associated with a use cover (e.g., soybeans crop, pasture, native forest, human-built structures).[3] Thus, changes in pixel classifications over time identify changes in land cover.

Measuring land cover— We construct a legacy forest stock ("as is") corresponding to all pixels in the satellite imagery occupied by forest cover in 1985 and 1986, the first years with available satellite data. Panel (a) in Figure 1 shows the legacy forest in the Amazon biome. Based on this initial forest stock, we track what happens to the cover in each legacy forest pixel over time. If a forest pixel becomes covered with patterns consistent with human action, we denote that pixel as deforested. Once a pixel is assigned to deforested, it can remain as that or become reforested. We classify an area as reforested if it stays deforested for at least two consecutive years and then is covered by forest for at least three years. This is done to avoid hyper-granular noise at the pixel level.

[3]Because our data spans several decades, different versions of Landsat were operational and therefore contributed with images to the classification scheme used by *MapBiomas*. More details on the classification methodology can be found at https://brasil.mapbiomas.org/wp-content/uploads/sites/4/2025/08/Amazon-Appendix-ATBD-Collection-10-v2.pdf.

Panels (b) through (d) in Figure 1 show the evolution of deforestation in the Amazon. By 2020, around 13% of the forest existing in 1985-1986 had been deforested, an equivalent of 52 million hectares (more than the area of California). A small area equivalent to 8 million hectares has been reforested, either by human action or by vegetation growth following abandonment.

This satellite imagery plays two important roles in the paper. First, it allows us to track deforestation at a very granular level and therefore quantify the environmental consequences of the Brazilian government's amnesty agenda. Second, it also enables us to directly establish human occupation of a particular area.

Rural parcels— Brazil has a centralized database attempting to organize records of rural parcels named Rural Environmental Property Registry (CAR). Rural property owners are required to submit information on their rural parcel, including several indicators meant to comply with environmental legislation and property boundaries to obtain a CAR code. The database is public and can be used by anyone to verify the status of a CAR code. The information submitted is self-reported and until Brazilian authorities verify the registration, the CAR code appears under the status "pending". However, because only an insignificant portion of submitted registrations is audited, effectively all parcels have pending CAR codes, which means legally and illegally occupied areas are indistinguishable in the database. This has created a loophole exploited by squatters, who submit registrations and obtain a (pending) CAR code, then use it informally as "proof" of occupation, especially for land transactions.

The CAR database therefore contains claimed property boundaries of legally and illegally occupied areas. While the nature of the data generating process is useful to us, because boundaries are self-reported, a large number of claimed property lines overlap. To avoid double-counting and other mechanical distortions, we develop a complex spatial resolution algorithm, described in Appendix C. Though implied magnitudes change, all our results carry if we used property boundaries without any adjustment (see Section 3.3.3).

In different parts of our analysis, we also use other sources containing property boundaries that have been audited or verified in different ways by Brazilian authorities. Those are described in detail in Appendix B.

Establishing human occupation— Land ownership claims in Brazil are mainly established through the idea of "ownership by use", where the most common strategy is to deforest portions of the land. This directly signals occupation to other potential land-grabbers and even fines received due to

illegal deforestation are used in future petitions for legalization, as environmental sanctions identify the squatter in the area.[4] Consequently, satellite evidence of deforestation in a particular area indicates human occupation, while a property claim in a forest area without any associated identifiable deforestation points to a CAR record that should be discarded.

Since it is not possible to perfectly observe the identity of squatters, we rely on a measure of deforestation to establish human occupation. Specifically, the first year when the share of deforested area in a rural parcel reaches 10% is assigned as the beginning of human occupation. We use 10% since for very small areas a lower share could be affected by noise in satellite readings, though marginally changing our threshold has little effect on our point estimates.

Figure 2 shows an example of the evolution of human occupation within a rural property of about 2,000 hectares. We track pixel-by-pixel the 24,000 satellite readings that span the property. In 2002, approximately 26% of the property was deforested, which meets our threshold to establish human occupation of 10%. By 2004, 38% of the property area had been deforested. By repeating the same process for each one of the hundreds of thousands of rural parcels in our data, we are able to measure eligibility for the amnesty policies and most of our outcomes.

3 The Consequences of the First Amnesty

intro paragraph

3.1 Empirical Design

Conditional on occupying land in a target area, the 2009 Land-Grabbing Amnesty sharply defines eligibility based on two conditions: having started the illegal occupation up to 2004 and having occupied no more than 1,500 hectares. This would suggest a straightforward difference-in-differences design where we would compare eligible squatters to those ineligible in target areas to estimate treatment effects of policy eligibility.

This difference-in-differences design, however, assumes that squatters who did not qualify for forgiveness provide a good counterfactual to those eligible. In practice, this rules out expectational effects,

[4] Even fencing, which is costly and without proper monitoring can be easily cut down or altered, requires forest cleaning prior to installation.

since land-grabbers assigned to the control group should behave as if the policy was absent.[5] Beyond proper empirical design considerations, any response from this untargeted group could be interpreted as unintended consequences from a policy design perspective and drastically affect net effects of the Amnesty. For example, if ineligible squatters interpret the 2009 policy as a signal of future forgiveness, they could respond by invading more land in target areas.

Setup— To jointly address the design issue above and study the response of ineligible squatters, we set up our empirical design in the following way. First, we use an appropriate control group of land-grabbers for which any forgiveness expectation is always non-existent and remains unchanged with the 2009 policy. We select land-grabbers in indigenous reservations and conservation areas, both federal land whose status is strictly regulated by law.[6] We refer to these group of land-squatters as *never-eligible* as they, in approximation, would never qualify to receive a title for the land they invaded.

Second, we consider a more general potential outcome function $Y_i(D_i^{\text{direct}}, D_i^{\text{expectations}})$, where a rural parcel i (or a "property" without a legal ownership title) with treatment assignment status receives one of the two treatments as follows. *Eligible* properties are exposed to the Amnesty direct treatment D_i^{direct} and satisfy the policy eligibility criteria:

$$\underbrace{\mathbb{1}\{i \in \text{target area}\}}_{\text{assignment}} \times \underbrace{\mathbb{1}\{\text{occupied}(i) \leq 2004\} \times \mathbb{1}\{\text{area}(i) \leq 1500\}}_{\text{eligibility}} \tag{1}$$

while *ineligible* properties are those in target areas exposed to an expectation treatment $D_i^{\text{expectations}}$ despite having invaded land after 2004, more area than 1,500 hectares, or both.

Characteristics of eligible, ineligible, and never-eligible land-grabbers— Table 1 shows a selected number of summary statistics for the two treated groups (eligible and ineligible properties) and

[5] Though not commonly discussed, the importance of expectational effects to proper identification of causal estimands is not exclusive to our setting. While many difference-in-differences implementations could credibly claim that the intervention was perceived as one-off, in principle any policy stipulating eligibility may generate anticipation of changes to these criteria at a future date. In cases of intertemporal changes in hazard of another intervention, like ours, it is crucial to identify a group for which under no set of conjectures eligibility status may likely change at a future date.

[6] The Brazilian Constitution prevents the introduction of any bill allowing the occupation of reservations by non-indigenous people (Article 231). Conservation units (e.g., national parks, biological reserves, ecological stations) are "specially protected territorial areas" whose suppression for private occupation would require the establishment of a complicated and overarching legal framework by Congress (Article 225). Besides the constitutional provision, a robust set of laws (e.g. Law 9985/2000) mandates expropriation of "private enclaves" from conservation units and also establishes that boundary reductions must each be made by a stand-alone, specific law. Consistent with this structure, land-use programs like the 2009 Land-Grabbing Amnesty have explicitly excluded these categories (Article 4 of the bill), leaving invaders with no reasonable path toward legalization.

the control group (never-eligible) we use to quantify the impact of the 2009 Amnesty. There are considerably more eligible properties, which are smaller than ineligible and also those never eligible. Eligible properties also tend to have higher shares of deforested area, adding up to 5.1 million hectares of deforested area in 2008. This is 24% more deforested area than that in ineligible properties and more than double of those in never-eligible parcels.

The first suggestive evidence of a strong response from ineligible land-grabbers around the policy comes from changes in deforestation. Table 1 shows that the deforested area in those properties not eligible for forgiveness increases by over 15% compared to a 11 percent-increase in the control areas. Between 2014 and 2009, an area larger than the US state of Delaware was deforested just in roughly 15,000 ineligible properties. Eligible properties display a much slower increase in deforested area, of about 6% or around 200 thousand hectares.

Main empirical model— Our main empirical design therefore compares in two separate regressions eligible and ineligible properties to never-eligible properties:

$$y_{it} = \beta x_i \times \mathbb{1}\{t \geq 2009\} + \upsilon_i + \upsilon_t + \varepsilon_{it} \tag{2}$$

where x_i is an eligibility indicator equal to one if rural parcel i satisfies the direct treatment assignment in (1). Equation (2) also includes parcel fixed effects, which control for unobservable heterogeneity across properties — e.g., differences in agricultural productivity or proximity to transportation routes — and year effects to absorb common shocks over time. In alternative specifications we use year-by-geography $\upsilon_t \times \upsilon_{g(i)}$ interactions (time effects interacted with state or municipality dummies). These capture time-varying characteristics at a local geographic area which control for differential changes in environmental enforcement, weather shocks, or local labor market trends. We cluster standard errors at the state level and use the data period of 2005 to 2014.

3.2 Impacts on Deforestation

We start our analysis of the consequences of the 2009 Land-Grabbing Amnesty on deforestation rates in eligible and ineligible properties.

Eligibility effects— Table 3 shows in columns (1) through (3) estimates on deforestation rates (deforested area$_t$ / total area$_t$) for rural properties eligible to apply for the amnesty. Results show that

program eligibility reduces deforestation rates by approximately 2%, or an estimated $\widehat{\beta}$ of 1 percentage point for an average deforestation rate of 58% in eligible properties before 2009. Panel (a) in Figure 4 plots event-study style estimates of the baseline model which show indistinguishable pre-trends in deforestation between eligible and control properties and a relative steady decline in the eligible group following the Amnesty law.

Expectation effects— In columns (4)–(6) of Table 3, we show treatment effects for properties ineligible for the first land-grabbing amnesty. Because squatters in these properties were not allowed the treatment intended by the law — legalization eligibility — any measured causal response could be interpreted as "unintended". This response could align with the stated goals of the policy, resulting in a decrease in deforestation as a positive spillover, for example, or counteract the effects from targeted squatters. Our results support the second conjecture, revealing a striking response: deforestation rates increase by over 30%, an effect one order of magnitude larger than the decrease in deforestation in eligible properties.

Net effects on deforestation— Using the reduced-form estimates above, we can compute the approximate average net effect of the first land-grabbing amnesty on deforestation in the Amazon. At the baseline, the 2.4% decrease in deforestation translates to a "counterfactual" deforestation rate 34% lower in eligible properties as a result of the amnesty.[7] In terms of forest cover, the average property avoided deforesting about 2 hectares:

$$\underbrace{-1.4}_{\widehat{\beta}\text{ (p.p.)}} \times \underbrace{58.4}_{\substack{\text{pre-2009}\\ \text{deforestation rate (\%)}}} \times \underbrace{69.03}_{\substack{\text{pre-2009}\\ \text{deforested area (ha)}}} \approx 2 \text{ ha}$$

We can then project the figure above to account for the total change in forest. Based on the numbers in Table 1, the total deforested area in eligible properties changed from 5.1 million hectares in 2008 to 5.27 million ha in 2014. This means that about 54,000 hectares of forest were kept in eligible properties as a result of the policy.

We can perform a similar exercise for ineligible properties, where amnesty induced a large increase in deforestation. Because ineligible areas tend to be larger than eligible properties, this difference in treatment effects amplifies to a 75 hectare "unintended" increase in deforestation compared to the 2

[7] The eligible property average deforestation rate increased from 58.4% to 62.5%, a 7% increase. Extrapolating, it would've been 9.4%, or 34% higher.

hectare decrease for the average eligible squatter:

$$\underbrace{4.2}_{\widehat{\beta}\text{ (p.p.)}} \times \underbrace{11.4}_{\substack{\text{pre-2009}\\ \text{deforestation rate (\%)}}} \times \underbrace{204.3}_{\substack{\text{pre-2009}\\ \text{deforested area (ha)}}} \approx 75 \text{ ha}$$

From the data, the deforested area increased to 4.66 million hectares in 2014 from 4.1 million ha in 2008. Since the average deforestation rate increased to 21.4% in the period, this implies that 42% of the observed deforestation in public land not targeted by the amnesty is attributable to expectation effects, an equivalent of 227,000 hectares of forest being cut. Taken together, intended and unintended responses to the amnesty resulted in a net removal of forest of 173 thousand hectares. This is a substantial number: on average, about 1% of the existing forest stock in non-designated areas is taken down every year, roughly 117 thousand hectares. The net effect of the amnesty thus amounts to almost 1.5 years of additional deforestation in a short timespan.

Equivalence in released carbon—The additional forest removed as a consequence of the 2009 Land-Grabbing Amnesty released a non-trivial amount of CO_2 into the atmosphere. Using the approximate 500 tons of CO_2 that the average hectare in the Amazon stores (**?**), the 173 thousand hectares of forest taken down we attribute as net effect released 86 million tons of CO_2, equivalent to 200 million barrels of oil burned (2 days of global oil use).

3.3 Robustness and Identification Assumptions

One paragraph detailing each subsection briefly.

3.3.1 Potential responses from land-grabbers in control areas

To interpret the previous estimated effects causally, our difference-in-differences design must assume away potential spillovers from treated to control units. This is because a change in the trajectory of deforestation rates in control areas as a result of the Amnesty would fail to provide a clean counterfactual to identify treatment effects. This could happen for example if land-grabbers in control areas decide to migrate to target areas, perhaps because they perceive the Amnesty law as signal of weaker enforcement in that area, or because they anticipate another amnesty in the future.

Whatever the underlying reason, testing for SUTVA (Stable Unit Treatment Value Assumption) violations is challenging, since we can never fully construct a counterfactual for control units without the actual occurrence of a policy. For example, consider the marginal potential land-grabber who would have invaded control areas but because of the policy, decides to invade land in target areas. We cannot observe data along this margin and, therefore, we cannot test for SUTVA violations of this type.[8] However, our setting provides us with data and variation to test for another margin which would clearly violate SUTVA.

Our test focuses on studying behavioral changes from individuals already present in one of the two areas (target and control) who migrate to the other. Particularly, consider individuals present in control areas migrating to target areas, equivalent to the conditional probability:

$$P(\text{target presence}_t = 1 \mid \text{control presence}_{t-k} = 1)$$

where k establishes an arbitrary previous period. The quantity above empirically matches to a difference-in-differences design that recovers the following estimand:

$$\gamma^{\text{SUTVA}} = P_{\text{post}}(m) - P_{\text{pre}}(m),$$

where $P(m)$ is the share of land-grabbers in target areas who moved from control areas. A value of $\gamma^{\text{SUTVA}} = 0$ indicates that the Amnesty policy did not induce differential migration out of control areas, consistent with a causal interpretation of our previous estimates. However, we do not have data that perfectly establishes presence of an individual over time and across space, leading to the following approach.

Environmental fines— We measure migration between target and control areas using data on environmental fines. Brazilian authorities make available rich, geolocated records of all fines related to environmental infractions in the country. Crucially for us, entities (individuals or firms) receiving a fine are identified by a unique tax number, allowing us to track repeat-offenders across space and time. Figure A.2 shows the spatial distribution of fines in target and control areas before and after the Amnesty.

[8] Note that the relevant margin for SUTVA violations involves the choice between target and control areas. This does not include an extensive margin of potential land-grabbers who decide to invade target areas or not invade at all, which is part of the effect we estimate in our deforestation results.

With data on environmental fines as a proxy for actual presence, the empirically relevant quantity becomes:

$$P(\text{control fined}_{t-k} = 1 \mid \text{target fined}_t = 1) \tag{3}$$

where we track whether a first-time infractor in target areas had previously been fined in control areas, therefore revealing migration.

Importantly, fines are only a noisy indicator of actual presence: they reveal presence only conditional on detection. Let π_t denote enforcement intensity, $P(\text{fined}_t = 1 \mid \text{presence}_t = 1)$, $m = 1$ an indicator for land-grabbers in target areas who moved from control areas and $s = 1$ an indicator for incumbent land-grabbers. Enforcement intensity can differ between these two groups, perhaps because migrant land-grabbers can better evade detection. Then, the following decomposition shows the bias introduced by using fines as a noisy indicator for presence:

$$\begin{aligned} P(\text{control fined}_{t-k} = 1 \mid \text{target fined}_t = 1) &= \frac{P(\text{control fined}_{t-k} = 1, \text{target fined}_t = 1)}{P(\text{target fined}_t = 1)} \\ &= \frac{\pi_{t-k}^{\text{control}} \times \pi_t^{\text{target}}(m) \times P(m)}{\pi_t^{\text{target}}(m) \times P(m) + \pi_t^{\text{target}}(s) \times (1 - P(m))} \end{aligned}$$

The expression shows the two sources of bias. The first is driven by enforcement intensity in control areas, which if varying over time, shifts the composition of individuals identified as "present in control". This could result for example in a declining share of fined movers even with positive true migration to control target areas. The second source of bias is the sample selection conditioning fined individuals in target areas. If enforcement rates between movers and incumbents differs, we only observe a biased sample in the target area — movers and incumbents are caught at different rates. If however we are willing to assume pooling, $\pi_t^{\text{target}} = \pi_t^{\text{target}}(m) = \pi_t^{\text{target}}(s)$, the decomposition above reduces to

$$P(\text{control fined}_{t-k} = 1 \mid \text{target fined}_t = 1) = \pi_{t-k}^{\text{control}} P(m)$$

so that simply controlling for enforcement in control areas completely removes the bias from inferring actual presence based on fines data.[9]

[9] We analyze the more general case without assuming pooling in Appendix D.

Empirical framework— We test for this SUTVA violation empirically with the following framework. For entry year t (defined as the first time individual or company i is fined in a target area), we create the outcome $\mathbb{1}\{\text{present in control}(i) < t\}$ which equals one if i was ever previously fined in the control, and run the following regression:

$$\mathbb{1}\{\text{present in control}(i) < \text{entry year in target}\}_t = \gamma \mathbb{1}\{\text{entry year}(i) \geq 2009\}_t + \varsigma_g(i) + u_{it} \tag{4}$$

where the regressor of interest is an indicator for post-policy years ($t \geq 2009$) in this baseline specification. A positive estimate of γ would suggest that individuals previously active in control areas began to reappear in target areas more often post-Amnesty — indicative of endogenous migration and hence, a SUTVA violation.

Taking differences between the post-policy and pre-policy periods and again assuming pooling enforcement in target areas yields the regression coefficient estimated by (4):

$$\widetilde{\gamma} = \pi_{\text{post}}^{\text{control}} P_{\text{post}}(m) - \pi_{\text{pre}}^{\text{control}} P_{\text{pre}}(m)$$

This interpretation of $\widetilde{\gamma}$ suggests that we directly control for environmental enforcement in control areas in model (4).

Estimating enforcement intensity— We measure enforcement intensity by leveraging a warnings system used by the Brazilian government called Real-Time System for Detection of Deforestation (DETER).[10] DETER runs high-frequency satellite monitoring that looks for changes in forest cover consistent with illegal deforestation. When the system detects a deforested area, it sends a warning alert which the authorities use to timely target enforcement efforts. The relative number of environmental fines for warnings issued therefore provides an empirical measure of enforcement intensity. This measure perfectly captures the source of bias in (4) if presence (associated with the deforestation) always results in a warning being issued. However, satellite-based warnings are sensitive to cloud coverage — presence is undetectable when cloud coverage prevents the visualization of the land underneath. To circumvent this technical issue, we follow **?** and adjust warnings using cloud coverage. Specifically, we

[10] We thank of Luis Maurano of Brazil's *Instituto Nacional de Pesquisas Espaciais* (INPE) for providing the data.

compute a cloud-adjusted enforcement intensity:

$$\widehat{\pi}_t^{\text{control}} = \frac{\text{fines}_t}{\text{warnings}_t} \times \%\text{ area visible}_t$$

which scales the measured enforcement intensity by the proportion of area without cloud coverage at a given point in time. This adjustment corrects imperfect monitoring based on the assumption that cloud patterns are exogenous and land-use behavior does not respond strategically to satellite observability. This assumption would be violated if for example land-grabbers decided to deforest more during cloudy periods.[11]

Results— Table 4 presents the results. Columns (1) and (2) show null estimates for changes in the probability of migration with municipality or state fixed effects. Column (3) controls for individual-specific moving propensity, which counts in how many different municipalities the same individual gets fined. This covariate intends to absorb to some extent correlated factors driving unobserved heterogeneity determining how good a land-grabber is at evading detection and also likely to migrate "anywhere" at the baseline. Adding this as a control increases the magnitude of the Amnesty dummy, but the estimate remains non-significant. In column (4), we report dynamic-style migration probabilities which are non-significant for all years relative to the pre-policy period.

Columns (5) and (6) of Table 4 include the enforcement intensity and cloud-adjusted enforcement intensity $\widehat{\pi}_t^{\text{control}}$, respectively. Results remain non-significant. Notably, in columns (7) and (8) we show that enforcement intensities change over time, underscoring the importance of controlling for this source of bias in our specifications. Taken together, we consider these estimates as supportive of no clear violations of the SUTVA margin we are able to test empirically — migration between target and control areas in response to the Amnesty.

3.3.2 Non-random exposure

Two paragraphs with a robustness to rule out bias introduced by non-random exposure.

[11] In Appendix D, we use run highly-local (30×30 meter) annual regressions of deforestation rates on number of cloudy days in that area and find that the two are uncorrelated. Note that because all our observations are at annual frequency, land-grabbers would need to put off deforesting for extended periods to exploit cloudy periods, since "delayed" satellite readings when the sky would eventually clear would still result in warnings being issued by DETER. If our identification relied on higher-frequency variation, this assumption would be less credible.

3.3.3 Alternative definitions and tests

Short paragraph detailing robustness (different occupation threshold, using CARs without the spatial cleaning etc) from table in Appendix.

4 Anticipating Another Amnesty

Our estimates in the previous section indicate that the increase in deforestation by land-grabbers ineligible for the 2009 Amnesty far exceeded conservation gains made from those eligible to receive a property title. We interpret the response from ineligible squatters as being driven by expectations effects of another amnesty in the future. In this section, we investigate the potential moral hazard introduced by the first policy in more detail.

Our results will show that, contrary to a simple signal of weaker environmental enforcement, land-grabbers finish

4.1 Internalizing "Acceptable" Levels of Land-grabbing

Consider a potential land-grabber after the 2009 Amnesty is passed on the margin of invading public land. Her eligibility status may change if a future amnesty allows for squatters with more recent occupation and if she complies with the area threshold set by the new policy. Conditional on the choice to invade, uncertainty over eligibility status is therefore primarily driven by the size of the public area she decides to occupy. Unbiased expectations over policy leniency would target the 1,500-hectare existing threshold, while specific stances on future leniency would result in different eligibility threshold sizes.

This suggests a simple test that tracks how much area land-grabbers decide to invade after the first amnesty eligibility criteria becomes known. We conduct this exercise in Figure 5. The figure shows how the cumulative share of land-grabbers in each group — ineligible and never-eligible — occupying parcels up to a certain size changes around the 2009 Amnesty. As the plotted response from never-eligible properties shows, the distribution of land grabbed in control areas is nearly identical after the first amnesty compared to before the policy was known. Moreover, the smooth, stable change in cumulative distributions also suggests that the 2009 Amnesty was not seen as an "across the board" weakening of

environmental enforcement, which could have stimulated land-grabbers to invade larger portions of land even in reservation and conservation areas.

In contrast to the control group, ineligible land-grabbers respond very strongly to the 2009 Amnesty. The share of squatters claiming a property area up to 1,500 hectares increases by 18 percentage points — an increase stable until the policy threshold. While part of this difference could be driven by composition changes in the sample of ineligible squatters, the sudden drop shown in the graph at the exact policy threshold suggests that eligibility criteria plays a major role in driving this response by squatters.

Holding fixed other time-varying unobservables, approximately 18 of land-grabbers had a response consistent with the expectation of another amnesty being at least as lenient as the first policy. Since the post-policy response stochastically dominates the pre-policy cumulative share for areas invaded as large as 5,000 hectares by several percentage points, a non-trivial portion of land-grabbers also behaved consistently with an expectation of more lenient future amnesty (which indeed occurred).[12]

This descriptive piece of evidence reinforces the role of the first land-grabbing amnesty in shaping expectations of ineligible land-grabbers over future potential amnesties.

4.2 Estimating Moral Hazard from Land Prices

We now move on to quantify the moral hazard introduced by the 2009 Amnesty using rural land prices. While it is straightforward to see why legalization offers an immediate premium to land eligible for amnesty — through the elimination of both eviction risk and the disamenity from trading in "gray" markets — the channel through which ineligible land would be affected, if any, may be less clear.

To fix ideas, assume that all agents expect that the 2009 Amnesty was one-off. Then, ineligible properties should not experience any direct price appreciation as their eligibility status will not change. But if agents assign some probability that another amnesty may occur, land prices of ineligible land will embed an option of future legalization. Depending on how much agents value a legal title to trade land

[12] It is important to stress that the shift toward more property size claims up to the 2009 Amnesty eligibility threshold does not imply less aggregate land-grabbing or a decrease in aggregate deforestation. As our results in Section 3 show, average deforestation rates and deforested areas by ineligible land-grabbers increase meaningfully after amnesty. Moreover, since these are claimed property boundaries and not based on deforested area, they measure exactly the margin that illegal occupants would target to comply with a future amnesty.

and their belief in a future amnesty, the Amnesty 2009 may indirectly create value to all land-grabbers in target areas, even those outside the eligibility scope of the policy.

We test how prices of eligible and ineligible land change due to the 2009 Amnesty by combining unique data on land prices in the Amazon based on with a conceptual model that backs out ...

Conceptual framework— Consider a market for land which can be occupied legally or illegally. Invaded public land is formally illegal but informally tradable. This means that defined property rights are valued by market participants, but not necessary for land to change hands. Let P_t^{legal} be the average market price of legal land at time t and P_t^{illegal} that of illegal land. We further assume that illegal land trades at a discount relative to legal properties, $P_t^{\text{illegal}} = P_t^{\text{legal}} - \omega_t$. The term ω_t can be interpreted as a legalization premium, capturing the value of not being subject to pecuniary and expected costs of enforcement, sanction, expropriation risk etc.

Before the Amnesty, eligible and ineligible properties are indistinguishably priced at P_t^{illegal}, since their eligibility status was deterministic. When the 2009 Amnesty goes into effect, parcels in public land gain a probability ϕ_t of legalization. Eligible parcels can apply and be legalized immediately and thus realize $\phi_t^{\text{eligible}} = 1$. Ineligible properties cannot be legalized under the terms of the new law. However, if agents believe there could be another amnesty in the future, they form $\phi_t^{\text{ineligible}} \leq 1$.

With the first amnesty, eligible parcels therefore experience an immediate price change of $\Delta_p^{\text{eligible}} = (P_t^{\text{illegal}} + \omega_t) - P_t^{\text{illegal}}$, so that ω_t is equivalent to the payoff to the option offered by a government amnesty. This option effectively represents a transfer from the government to land-grabbers which gets captured in property values. Ineligible properties, in contrast, receive a dynamic option capturing the value of becoming legalized at some future point:

$$P_{t\geq 2009}^{\text{ineligible}} = P_t^{\text{illegal}} + \mathbb{E}_t\left[e^{-r\tau}\phi_t\omega_t\right] = P_t^{\text{legal}} - \omega_t\left(1 - \phi_t e^{-r\tau}\right)$$

where τ is the expected time until renewed amnesty and ϕ_t can be interpreted in this formulation as the belief in future policy renewal.

Empirical design— Let (g,t) index a geographic area in a given year. The average log price of transacted land is given by p_{gt} and combines sales of both legal and illegal land.[13] Splitting illegal

[13]The sampling of both legal and illegal land is a useful feature in our price data source. Rather than relying exclusively on public deeds, the data is based on surveys hundreds of land market participants, including banks, brokerage houses,

properties into eligible and ineligible for the 2009 Amnesty, log land prices can be expressed as follows:

$$p_{gt} = \log\left(p_{gt}^{\text{legal}} s_{gt}^{\text{legal}} + p_{gt}^{\text{eligible}} s_{gt}^{\text{eligible}} + p_{gt}^{\text{ineligible}} s_{gt}^{\text{ineligible}}\right)$$

where s_{gt} represents the share of that type of land in total sales which adds up to one. Due to the nature of our data, we cannot observe the makeup of turnover at a given point in time. Instead, we compute shares of each property type in a given geographic area $\widetilde{s}_g^k = \sum_{i \in g} \mathbb{1}\{i = k\}$ and treat those as propensities to trade. The intuition here is as follows: if 80% of rural properties in a certain area are legally-occupied, transactions in that area are more likely to incorporate legal land. We assess the plausibility of this intent-to-treat style assumption shortly below.

We then estimate the following two-way fixed effects model:

$$\log P_{gt} = \alpha + \gamma_1 \widetilde{s}_g^{\text{eligible}} \times \text{post}_t + \gamma_2 \widetilde{s}_g^{\text{ineligible}} \times \text{post}_t + \lambda_t + \mu_g + \varepsilon_{gt} \tag{5}$$

where post_t is an indicator for the 2009 Amnesty and afterwards. We run this regression at the level of the geographic regions used in the data, equivalent to a mix of micro- and meso-regions in the Amazon. For each region, we identify all eligible, ineligible, and legal rural properties and calculate the shares of each type of property from the total counts in that region. We are able to match 26 regions with prices spanning 2004 to 2014.

Since the category omitted in (5) denotes properties occupied legally, the coefficients γ_1 and γ_2 measure the post-Amnesty price change in areas with a higher concentration of properties eligible for forgiveness and those ineligible, respectively. Based on the real options framework we introduced above, we can give a structural interpretation to these parameters.[14]

agricultural associations, farmers, and local municipalities on recent sales for rural land. More details can be found in Appendix B.

[14] We use properties already legalized as the main control in this analysis, as opposed to land-grabbers in reservation and conservation areas as we did in Section 3, to completely shut-off expectation effects from the control group. not sure whether to have this footnote here or not.

To write out the first-difference estimate, we use the assumption that pre-policy prices of eligible and ineligible land were identical to that of illegal land:

$$\begin{aligned}
p_{gt}^{\text{pre}} &= \log\left(s_g^{\text{legal}} P_{gt}^{\text{legal}} + s_g^{\text{eligible}}(P_{gt}^{\text{legal}} - \omega_t) + s_g^{\text{ineligible}}(P_{gt}^{\text{legal}} - \omega_t)\right) \\
&= \log\left(P_{gt}^{\text{legal}}(s_g^{\text{legal}} + s_g^{\text{eligible}} + s_g^{\text{ineligible}}) - \omega_t(s_g^{\text{eligible}} + s_g^{\text{ineligible}})\right) \\
&= \log\left(P_{gt}^{\text{legal}} - \omega_t(s_g^{\text{eligible}} + s_g^{\text{ineligible}})\right)
\end{aligned}$$

The above implies that the marginal effects for s_g^{eligible} and $s_g^{\text{ineligible}}$ are the same and equal to:

$$\frac{\partial}{\partial s_g^{\text{eligible}}} p_{gt}^{\text{pre}} = \frac{\partial}{\partial s_g^{\text{ineligible}}} p_{gt}^{\text{pre}} = -\frac{\omega_t}{P_{gt}^{\text{legal}} - \omega_t(s_g^{\text{eligible}} + s_g^{\text{ineligible}})}$$

Similarly, we can rewrite the average log price post-Amnesty as:

$$\begin{aligned}
p_{gt}^{\text{post}} &= \log\left(s_g^{\text{legal}} P_{gt}^{\text{legal}} + s_g^{\text{eligible}} P_{gt}^{\text{legal}} + s_g^{\text{ineligible}}[P_{gt}^{\text{legal}} - \omega_t(1 - \phi_t e^{-r\tau})]\right) \\
&= \log\left(P_{gt}^{\text{legal}}(s_g^{\text{legal}} + s_g^{\text{eligible}} + s_g^{\text{ineligible}}) - s_g^{\text{ineligible}} \omega_t(1 - \phi_t e^{-r\tau})\right) \\
&= \log\left(P_{gt}^{\text{legal}} - \omega_t s_g^{\text{ineligible}}(1 - \phi_t e^{-r\tau})\right)
\end{aligned}$$

which means that legal and eligible properties become homogeneous for identification purposes. Thus, only ineligible parcels are discounted as traded land, so that the post-policy marginal effect is

$$\frac{\partial}{\partial s_g^{\text{ineligible}}} p_{gt}^{\text{post}} = \frac{-\omega_t(1 - \phi_t e^{-r\tau})}{P_{gt}^{\text{legal}} - \omega_t s_g^{\text{ineligible}}(1 - \phi_t e^{-r\tau})}$$

The coefficients of interest γ_1 and γ_2 can be interpreted structurally by subtracting post-Amnesty marginal effects from pre-Amnesty effects:

$$\gamma_1 = 0 - \left(-\frac{\omega_t}{P_{gt}^{\text{legal}} - \omega_t(s_g^{\text{eligible}} + s_g^{\text{ineligible}})}\right) = \frac{\omega_t}{P_{gt}^{\text{legal}} - \omega_t s_g^{\text{illegal}}}$$

which picks up the relative price improvement in eligible land relative to legal land due to the policy. We can finally write γ_1 as:

$$\gamma_1 = \frac{\omega_t}{P_{gt}^{\text{legal}}} \frac{1}{1 - \frac{\omega_t}{P_{gt}^{\text{legal}}} s_g^{\text{illegal}}} = \frac{\omega_t}{P_{gt}^{\text{legal}}}(1+\delta_1) \tag{6}$$

where $\delta_1 \equiv s_g^{\text{illegal}} \omega_t / P_{gt}^{\text{legal}}$, a correction term that measures how much illegal land "biases" the treatment effect for eligible land. Similarly, we can compute γ_2:

$$\begin{aligned}
\gamma_2 &= \frac{-\omega_t(1-\phi_t e^{-r\tau})}{P_{gt}^{\text{legal}} - \omega_t s_g^{\text{ineligible}}(1-\phi_t e^{-r\tau})} - \left(-\frac{\omega_t}{P_{gt}^{\text{legal}} - \omega_t(s_g^{\text{eligible}} + s_g^{\text{ineligible}})} \right) \\
&= \frac{\omega_t [P_{gt}^{\text{legal}} + \omega_t s_g^{\text{ineligible}}(\phi_t e^{-r\tau} - 1) + (P_{gt}^{\text{legal}} - \omega_t s_g^{\text{illegal}})(\phi_t e^{-r\tau} - 1)]}{(P_{gt}^{\text{legal}} - \omega_t s_g^{\text{illegal}})[P_{gt}^{\text{legal}} + \omega_t s_g^{\text{ineligible}}(\phi_t e^{-r\tau} - 1)]} \\
&= \frac{\omega_t}{P_{gt}^{\text{legal}}} \phi_t e^{-r\tau}(1+\delta_2) \qquad (7)
\end{aligned}$$

where $\delta_2 \equiv \omega_t / P_{gt}^{\text{legal}}(\phi_t e^{-r\tau} s_g^{\text{ineligible}} + s_g^{\text{eligible}})$.

With the structural interpretation for the coefficients of (5), we define $\gamma_1 - \gamma_2$ as a "recidivism" wedge, i.e., the difference between the value of immediate legalization and speculative value (moral hazard) of ineligible land. It is straightforward to see that as $\phi_t e^{-r\tau} \longrightarrow 1$, i.e., market participants believe the eligibility status of ineligible land is likely to change due to a future amnesty, $\gamma_2 \longrightarrow \gamma_1$.

Results— Table 5 shows the results. We first discuss the estimates of γ_1 and γ_2. Column (1) runs model (5) on a shorter window around the 2009 Amnesty, from 2006 to 2011. Estimates show a sharp and immediate price response in regions with larger shares of eligible and ineligible land. Column (2) cover the full sample period for our data, similar to our main empirical results in Section 3. Point-estimates for both γ_1 and γ_2 become larger, with the immediate legalization premium to eligible parcels being higher than the option value for ineligible land. This implies a relatively small recidivism wedge of only 1-percent growth in land prices for eligible properties compared to those whose eligibility status could change in the future. In practice, the land market seems to have almost entirely priced in another amnesty.

Composition changes and turnover— Our empirical implementation uses the distribution of property types within each region as propensities to trade. That is because we cannot observe parcel-

level transactions and as a consequence the composition of sales. We can assess to some extent the plausibility of this assumption leveraging once again the environmental infractions data we used in Section 3.3.1. Due to the empirical limitations of how fines actually track underlying presence, we the following exercise as more suggestive in nature.

When properties are fined because of environmental violations, property occupants are named in the infraction records. Therefore, if we observe the same parcel being fined more than once, we can establish occupation patterns. Specifically, classify a parcel whose occupant changes for the first time in year t, without having the previous occupant reappearing in the fines database, as a land sale in that year. We then adjust raw counts of each group type (legal, eligible, and ineligible) by the matching rate we achieve when merging the geolocated fines data with the parcel boundaries from the CAR database. This approach yields annual shares of each property type in that year rural land turnover.

Columns (3) and (4) show results of predictive-style regressions of the share of eligible and ineligible properties in turnover, respectively, on the eligible and ineligible shares in the distribution of rural parcels. We also include year fixed effects in these regressions. Estimates show a strong association between the shares used to estimate γ_1 and γ_2 and the noisy proxy for turnover.

Moral hazard computation— Our real options framework allows us to back out two important quantities to evaluate the moral hazard introduced by the first land-grabbing amnesty. First, we can compute the approximate average premium of legalization received by land-grabbers eligible for the amnesty. Solving for $\widehat{\omega}_t$ in (6) and using the average land price as of 2008 in regions with only legal properties and the average share of illegally-occupied land, we obtain:

$$\widehat{\omega}_t \approx \frac{\widehat{\gamma}_1 \overline{P}_t^{\text{legal}}}{1+\widehat{\gamma}_1 \overline{s}^{\text{illegal}}} = \frac{0.518 \times \$681}{1+0.518 \times 57\%} = \$272/\text{ha}$$

where $\overline{P}$ and $\overline{s}$ are averages across all geographic regions. The estimate represents a 40% land value direct transfer from the Brazilian government to eligible land-grabbers, an aggregate value generation of over 2.8 billion BRL (approximately \$1.6 billion at the end of 2009).

Similarly, we can compute the implied probability of another amnesty at the sample means and using $\widehat{\omega}_t$ and $\widehat{\gamma}_2$:

$$\widehat{\phi_t e^{-r\tau}} = \frac{\widehat{\gamma}_2 \left(\overline{P}_t^{\text{legal}} - \widehat{\omega}_t \overline{s}^{\text{eligible}} \right)}{\widehat{\omega}_t \left(1 + \widehat{\gamma}_2 \overline{s}^{\text{ineligible}} \right)} = \frac{0.405 \times (\$681 - \$272 \times 46\%)}{\$272 \times (1 + 0.405 \times 46\%)} = 79\%$$

We estimate that the post-policy option value of ineligible land reflects a present-discounted belief in future legalization of 79%. This suggests that the market substantially internalized expectations of another amnesty, which led prices of areas with more ineligible land to appreciate to a large extent as if these properties had already been offered legalization.

4.3 Winners from Another Amnesty

By expanding eligibility thresholds, the 2017 Land-Grabbing Amnesty likely rewarded public land invasions that occurred after the first amnesty. During the 2009 Amnesty, 70% of ineligible land-grabbers failed to comply with the start of occupation threshold. While some of these land-grabbers invaded land during 2005 and 2008, many grabbed land after the first amnesty went into effect. The expansion of eligibility criteria in the 2017, making invasions until 2011 and with areas up to 2,500 hectares eligible, changed the eligibility status of 53% previously ineligible parcels. 35% of them includes land grabbed after 2009.

Our previous estimates indicate that a land-grabbing amnesty provides infractors with a large monetary reward through an increase in land prices, whether they are directly targeted or not. This expectations channel implied a strong belief by ineligible squatters that the land they would take would eventually receive another amnesty, and over a third of them would have had correct beliefs vis-à-vis the 2017 Amnesty.

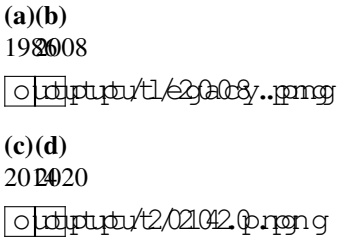
This dynamic is captured in column (5) of Table 5 where we check whether areas with larger land price changes after the first amnesty also higher land-grabbing participation by invasions after 2009 that would become eligible for the second amnesty. We find a 6.8% price increase associated with a 10 p.p. higher share of "to-become" eligible properties that were occupied after, and presumably in response to, the first amnesty.

5 Conclusions

References

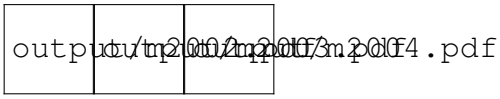
Figures and Tables

FIGURE 1: DEFORESTATION EVOLUTION IN THE AMAZON



NOTES— These are plots generated with satellite 30×30-meter resolution pixels of land coverage for the Amazon biome in Brazil. Plot **(a)** shows "legacy forest" stock, measured by pixels corresponding to forest cover in 1985 and 1986. The "other" category are all non-forest pixels, including pre-existing deforestation and non-natural land cover. Panels **(b)** to **(d)** show the evolution of the legacy forest stock along two margins: deforestation and reforestation. We call a pixel deforested for the first time in year t if it transitioned from a forest cover in $t-1$ to one with cover consistent with human action, and reforested if it had a two-year period as deforested followed by three years with vegetation coverage. We adopt forest and human cover definitions from MapBiomas, from where we also use the land cover classification methodology and output. More details in Appendix B.

FIGURE 2: ESTABLISHING HUMAN OCCUPATION WITH SATELLITE DATA



NOTES— Land use derived from satellite imagery for a rural property (based on CAR) in Rondonia state. Self-reported property boundaries are shown in blue. The deforestation rate increased from 26% to 38% during 2002-2004, corresponding to about 824 hectares. Raw image resolution of a plotted pixel is 30×30 meters. This property spans over 24,000 pixels.

FIGURE 3: ELIGIBILITY ASSIGNMENT — 2009 LAND-GRABBING AMNESTY

(a) Step 1: Target Areas

(b) Step 2: Eligible Properties

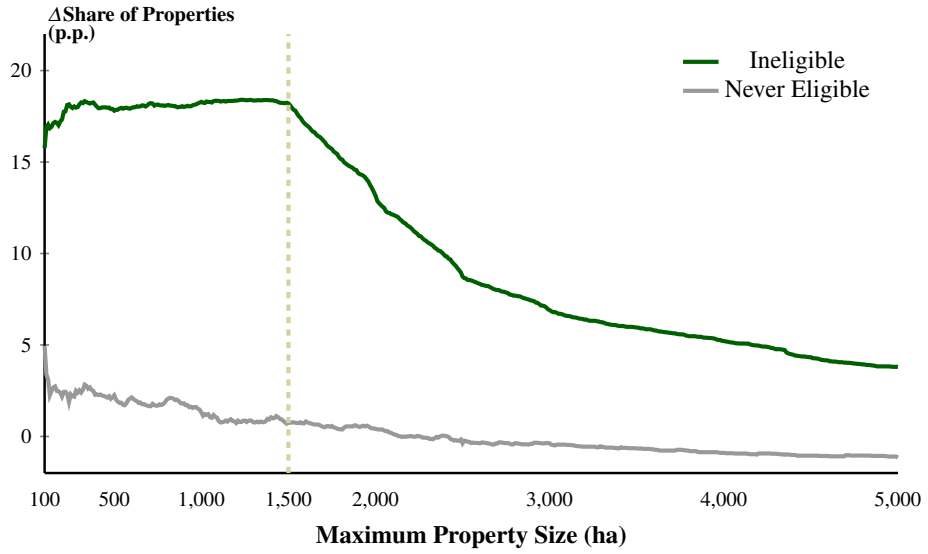
output/fig1_final.pdf output/fig2_final.pdf

NOTES— The figure shows rural properties by treatment status determined by Brazil's 2009 Land-Grabbing Amnesty. Panel **(a)** displays **target areas** and **those excluded** from amnesty eligibility. Only undesignated federal land was targeted in the amnesty. State-owned land, forestland allocated to indigenous reservations, conservation units, or land designated to any other use were excluded from the policy. Federal land allocated to reservations and conservation as the control areas. Panel **(b)** shows rural parcels within target areas that were eligible and ineligible for amnesty, as well as never-eligible parcels in control areas. **Eligible** rural properties are those with occupation prior to 2005 and smaller than 1500 hectares. Other illegal occupations in the target areas not complying with these thresholds are **ineligible**. Properties in the control areas are **never eligible**. Property claim boundaries plotted have been corrected using the algorithm described in Appendix B. See main text for more details on treatment assignment and the 2009 Amnesty.

FIGURE 4: ELIGIBILITY AND EXPECTATION EFFECTS ON DEFORESTATION — 2009 LAND-GRABBING AMNESTY

NOTES— These are event-study plots of the form $y_{it} = \upsilon_i + \upsilon_t + \sum_{r \neq -1} \beta_r \times x_i \times \mathbb{1}\{t = r+1\} + \varepsilon_{it}$, where x_i denotes treatment eligibility of rural property i, $\mathbb{1}\{i \in \text{target area}\} \times \mathbb{1}\{\text{occupied}(i) \leq 2004\} \times \mathbb{1}\{\text{area}(i) \leq 1500\}$, for the 2009 Land-Grabbing Amnesty. Panel **(a)** shows treatment effects for deforestation rates comparing properties eligible for the Amnesty to those never-eligible (located in indigenous and conservation lands). These are dynamic effects of the point-estimate in column (1) of Table 3. Plot **(b)** shows treatment effects for the same outcome but with the treatment indicator for ineligible properties, which are then compared to the never-eligible group (dynamic effects for column (4) in Table 3). Error bars are 95% confidence intervals computed with standard errors clustered at the state level. Deforestation rates are adjusted for forest regrowth (reforestation) or abandonment.

FIGURE 5: ANTICIPATING ANOTHER AMNESTY — CHANGES IN INVADED PROPERTY AREAS



NOTES— These curves report the change in the cumulative share of land-grabbing up to a certain property size before and after the 2009 Land-Grabbing Amnesty. We calculate each point of each curve as the difference between the proportion of land-grabbers entering after the Amnesty (2009-2014) and before (2005-2008) who claimed a property area greater or equal than an area. For example, land-grabbing of properties up to 1,500 hectares increases 18 p.p. after the Amnesty is introduced. **Ineligible** properties are those that did not qualify for amnesty, either because the land was occupied after 2004 or exceeded 1,500 hectares. **Never eligible** properties are located in reservation or conservation areas. The **dashed line** marks the maximum property size eligible for amnesty. We censor the *x*-axis at 5,000 hectares for better visibility.

TABLE 1: CHARACTERISTICS OF OCCUPATIONS IN TARGET AND CONTROL AREAS

	Eligible	**Ineligible**	**Never eligible**
# Properties	71,171	15,254	7,049
Property Area (ha, $t < 2009$)	143	661	760
Property Deforestation Rate (ha, $t < 2009$)	58.4%	11.4%	35.7%
Δ Property Deforested Area	6.3%	15.6%	11.5%
Total Deforested Area (millions of ha)			
$t = 2008$	5.1	4.1	2.0
$t = 2014$	5.3	4.7	2.2

NOTES— These are selected characteristics of rural parcels with different eligibility status during the 2009 Land-Grabbing Amnesty. Eligible properties are those in target areas whose occupation began before 2005 and is less than 1,500 hectares. Ineligible properties are also located in target areas, but do not comply with the amnesty eligibility thresholds. Never eligible properties are located in control areas (indigenous reservations and conservation areas). Until 2008, all of these rural parcels illegally occupied public land in the Amazon. We define deforestation rates as the share of a property claim's area that has been deforested. Property areas are calculated directly from boundaries submitted by land occupants.

TABLE 2: 2009 AMNESTY TAKE-UP AND ELIGIBILITY

	Applies for Amnesty		Receives Amnesty	
	(1)	(2)	(3)	(4)
% Deforested Area	0.047**	0.043*	−0.171	−0.022
	(0.019)	(0.018)	(0.193)	(0.225)
% Deforested Area^2	−0.052**	−0.057**	0.167	0.070
	(0.018)	(0.023)	(0.149)	(0.163)
$\mathbb{1}\{\text{area} = ...\}$				
≤ 100	0.080***	0.066***	0.103**	0.099**
	(0.007)	(0.014)	(0.043)	(0.032)
$(100, 500]$	0.062***	0.053***	−0.054	−0.068*
	(0.008)	(0.011)	(0.058)	(0.034)
$(500, 1000)$	0.038***	0.032***	−0.118	−0.117
	(0.010)	(0.008)	(0.106)	(0.090)
Occupation time	−0.009	−0.008	−0.031**	−0.039***
	(0.015)	(0.150)	(0.010)	(0.009)
Occupation time^2	0.0001	0.0001	0.001**	0.001***
	(0.0004)	(0.0001)	(0.0003)	(0.0003)
Fixed Effects				
State				
Outcome Baseline	10.2%		42.2%	
Observations	73,809		7,507	

NOTES— This table shows application rates (columns (1) and (2)) and amnesty rates, conditional on applying (columns (3) and (4)) of land-grabbers eligible in the 2009 Land-Grabbing Amnesty. In each model, we regress an indicator on the following parcel characteristics: % of deforested area in 2008, property size bins (larger than 1000 hectares is the omitted category), and occupation time until 2008 (defined as the first year since 1989 when that property shows at least 10% of forest being taken down). Even columns also include state fixed effects. All specifications report standard errors clustered at the state-level.

TABLE 3: 2009 AMNESTY ELIGIBILITY AND EXPECTATION EFFECTS

	Deforestation Rate					
	(1)	(2)	(3)	(4)	(5)	(6)
Amnesty Eligibility	−1.412**	−0.844**	−0.829**			
	(0.558)	(0.325)	(0.304)			
Amnesty Expectation				4.204***	4.047***	3.744***
				(0.886)	(0.815)	(0.662)
Fixed Effects						
Property+Year						
Property+Year×State						
Property+Year×Municipality						
Outcome Baseline		58.4%			11.4%	
Estimated Effect (%)	−2.4%	−1.4%	−1.4%	36.9%	35.5%	32.8%
Observations		782,175			231,833	

NOTES— This table shows treatment effect estimates from the difference-in-differences specification $y_{it} = \beta x_i \times \mathbb{1}\{t \geq 2009\} + \upsilon_i + \upsilon_t + \varepsilon_{it}$, where $x_i = \mathbb{1}\{i \in \text{target area}\} \times \mathbb{1}\{\text{occupied}_i \leq 2004\} \times \mathbb{1}\{\text{area}_i \leq 1500\}$ assigns treatment eligibility to the 2009 Land-Grabbing Amnesty in Brazil. Control units are never-eligible properties in indigenous and conservation lands. Columns (1) and (4) implement the baseline two-way fixed effect specification with property and year dummies. Columns (2) and (5) include property and state-by-year fixed effects, and columns (3) and (6) contain property and municipality-by-year fixed effects. Standard errors in all specifications are clustered at the state-level. The outcome baseline is measured as the average prior to 2009 for treated properties, which we then use to compute percentage-equivalent estimated effects.

TABLE 4: ASSESSING THE RESPONSE FROM LAND-GRABBERS IN CONTROL AREAS

	Previously Fined in Control						Enforcement Intensity	
							Target Areas	Control Areas
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\mathbb{1}\{\text{entry year} \geq 2009\}$	0.002	0.001	0.002		0.004	0.003		
	(0.002)	(0.001)	(0.002)		(0.002)	(0.002)		
$\mathbb{1}\{\text{entry year} = ...\}$								
2006				0.000				
				(0.003)				
2007				−0.001				
				(0.003)				
2009				0.001				
				(0.004)				
2010				0.005				
				(0.003)				
2011				0.002				
				(0.003)				
2012				0.003				
				(0.003)				
2013				0.002				
				(0.004)				
2014				−0.001				
				(0.002)				
$\mathbb{1}\{t \geq 2009\}$							0.033	0.031**
							(0.018)	(0.012)
Fixed Effects								
Municipality								
State								
Controls								
Mover propensity								
Enforcement intensity								
Adjusted enforcement intensity ($\widehat{\pi}_t^{\text{control}}$)								
Outcome Baseline								
Before 2009			0.16%				1.38%	0.50%
After 2009			0.32%				4.67%	3.59%
Observations			5,655				9	

NOTES— This table conducts several tests for potential violations of the Stable Unit Treatment Value Assumption (SUTVA). All model specifications rely on data combining: geolocated environmental fines, warnings issued by the Real-time System for Detection of Deforestation (DETER), and cloud coverage in the Amazon. Columns (1) and (2) implement the model $\mathbb{1}\{\text{present in control}(i) < \text{entry year in target}\}_t = \gamma\mathbb{1}\{\text{entry year}(i) \geq 2009\}_t + \varsigma_g(i) + u_{it}$ with ς_g using municipality and state dummies, respectively. Column (3) implements the same model with municipality fixed effects and with mover propensity as a covariate. We construct this variable by tracking whether the same squatter is fined in different municipalities over time. Column (4) implements this last model using event-study style estimates based on the year of entry (instead of the dummy for the post-2009 period). Columns (5) and (6) control for imperfect observation of presence using fines by adding enforcement intensity as a covariate. Column (5) uses an unadjusted measure of intensity, $\text{fines}_t/\text{warnings}_t$, while column (6) uses the adjusted intensity $\widehat{\pi}_t^{\text{control}}$ we define in the main text. This adjustment uses the % of area visible at a given point in time when deforestation warnings are issued using satellite data. Columns (7) and (8) test for changes post-Amnesty of this enforcement intensity measure in target and control areas. All models through (6) cluster standard errors at the municipality level. Models (7) and (8) have robust standard errors.

TABLE 5: MORAL HAZARD ESTIMATES USING LAND PRICES

	Land Price		% in Turnover		Price Change
			Eligible	Ineligible	
$t =$	$[2006, 2011]$	$[2003, 2014]$	$[2003, 2014]$		$\overline{[2009, 2014]} - \overline{[2004, 2008]}$
	(1)	(2)	(3)	(4)	(5)
% Eligible properties	0.419**	0.518***	0.792***		
	(0.144)	(0.108)	(0.076)		
% Ineligible properties	0.344***	0.405**		0.797***	
	(0.074)	(0.156)		(0.242)	
% To-become eligible properties					0.681**
					(0.333)
Fixed Effects					
Region					
Year					
Observations		312			28

NOTES— Columns (1) and (2) show log land price regressions in a region on the share of occupations that were eligible and ineligible for the 2009 Land-Grabbing Amnesty. The omitted category are already legalized parcels. These specifications have region and year fixed effects and standard errors clustered by year. A region is defined in the land price dataset and matches approximately a micro- or macro-region in the Brazilian Amazon. Column (1) runs the model for 3 years before and 3 years after the Amnesty is in effect. Column (2) runs for the full period. Columns (3) and (4) regress a constructed measure of the participation of each eligibility type in annual sales on the shares used in the first two columns. These shares are propensities to sell, whereas the outcomes in columns (3) and (4) are noisy measures from sales using different individuals being fined in the same property over time, suggesting change in occupation (a sale). Column (5) compares the change in log average prices in regions where higher shares of ineligible squatters (in the 2009 Amnesty) would become eligible for the 2017 Amnesty and that invaded land after the first Amnesty.

Appendix A Figures and Tables

FIGURE A.1: 2009 LAND-GRABBING AMNESTY TAKE-UP

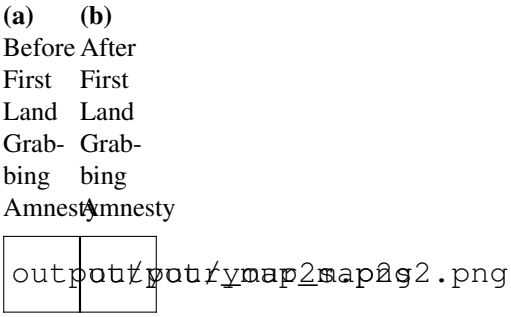
(a) Number of Amnesty Requests

(b) Total Area Requested (thousands of hectares)

output/output/policy_1c_yr1e_request_sp.pdf

NOTES— These plots show official records with applications submitted to the office in charge of the 2009 Land-Grabbing Amnesty rollout. We plot the annual number of requests (Panel **(a)**) and the total area combined (Panel **(b)**)

FIGURE A.2: SPATIAL DISTRIBUTION OF ENVIRONMENTAL FINES IN THE AMAZON



NOTES— This figure shows environmental fines in target and control areas accumulated before and after the 2009 Land-Grabbing Amnesty. Environmental fines include a host of infractions and we use them as a proxy for presence in a certain localized area in different parts of the paper.

Appendix B Data and Variable Definitions

1. Keep CARs in the Amazon biome
2. CARs with more than 0.1% of their area inside a federal gleba are "dirty", the others "clean"
3. For CARs outside federal glebas, find out those not in state glebas and inside reservation or indigenous areas
4. CARs outside federal and state glebas, and outside reservation or indigenous areas are "pure control"

1. Step 1: Define sample of CARs "ever occupied" (up to 2014)

A Identifying irregular land-use with CAR, TerraBrasilis and CNFP

We identify irregular land-use in the Amazon biome with the help of geospatial land-use data from the several sources. Our output data-set is structured at the municipality-year level and identifies the amount of sensitive government ear-marked land which has been claimed by individuals. We begin with self-reported and partially verified CAR (Cadastro Ambiental Rural) rural land demarcations and compute the area of intersection (m^2) of these land claims with officially recognized indigenous, conservation and Federal forested lands, taken from TerraBrasilis and CNFP. Our data-set spans 2013-2023 and includes municipalities for the entire Amazon Biome and all of the CARs within the biome.

A.1 Canceled CARs

Of the 892,670 CARs in the Amazon biome, 32,553 or roughly 3.5% of are listed as 'Cancelled'. Disconsidering any overlaps, this figure amounts to 11% of the area of all CARs in the same territory. CAR cancellation is an administrative process undertaken by Sicar, however we still have limited information about the underlying motivations and processes.

Figures **??** and **??** illustrate that CAR cancellation is not only a function of: 1) when a land claim is made, or 2) whether a claim intersects with other claims. Specifically, in these figures, we see five separate CAR claims that appear in chronological order, most of which overlap, but only two of which have been cancelled, namely the first and last CAR claims. For simplicity, call these claims A, B, C, D and E. Claims A and E have been canceled, claims B and D are active and claim C is pending. Claim A intersects with all other claims. Claim B intersects with all others barring E. Claim D completely engolfs claims B and E. Clearly, CAR cancellation depends on other rules which we are currently unaware of and are likely linked to official land purchases.

Figures **??**, **??** and **??** map the distribution of canceled CARs both in absolute terms and as a share of the total number of CARs in each municipality. We observe that as the number of CARs increases, so does the percentage of cancellations, implying a non-linear relationship in cancelation rates. Moreover, in **??**, we see that there are some municipalities with particularly large numbers of cancellations; useful information for us to probe even further if we wish to.

Cancelled CARs also have their information stripped from the official data. In order to recuperate details like "Registry Date", to gauge when the land was initially claimed, we purchase the information from a third-party data-broker (Registro Rural). Below, we present summary statistics about these canceled CARs and map their distribution.

B CAR Intersections

It is not uncommon for property claims to overlap in the Amazon[15]. Of the 829,260 CARs we have processed, 86% intersect with at least one other CAR. On average, conditional on having at least one intersection, a CAR overlaps 3.6 times, with each intersection composing roughly 22.73% of its area. Canceled CARs have even higher levels of intersections with other CARs. In our sample, we observe 29,437 canceled CARs. Of these, 93.17% have at least one intersection and overlap with other CARs 5.54 times. Each intersection is on average 24.8% of the canceled CAR area.

There does not appear to be any significant temporal relationship which can be exploit to clearly explain the reason for a CAR cancellation. This holds both over time and relative to conflicting CARs. Overall, there are: 1,380,510 unique CAR conflicts in our data. Of these, the share of CAR conflicts where a reference CAR is canceled/pending/suspended/active is: 5%, 27%, 3%, and 65%. For CARs which are eventually canceled, 74% of the created CARs have at least one conflict. This figure is similar for pending, suspended and active CARs: 66%, 76%, 69%.

C Irregular land-use variables at the municipal level

The main irregular land use variables we construct aim at estimating the amount of protected land that is claimed by individuals. We consider three types of protected lands: indigenous, conservation and demarcated forested lands and compute the variables by computing the area of intersection between these variables and the planar union of all CARs—active and inactive. Since the individual CAR property geometries are stored in municipality level shape-files, we read in each municipality m shape-file at a time perform the following cleaning steps:

1. Convert all Simple Features (SF) object to CRS=4674; this includes the three sources of protected land-use
2. Convert invalid polygons into valid polygons
3. Take the union of all CAR polygons (henceforth UCAR)
4. Compute UCAR area to determine the baseline claimed area
5. Compute the area of intersection of indigenous, conservation, and forested; segregating forested land into three types: A, B and C.
6. Take the union of cancelled (UCCAR) and remaining CARs (URCAR), separately
7. Compute the areas of UCCAR and URCAR individually and their intersection

Since all CARs have associated registration dates, we perform the above process for each year in the data-set, adding each year's new CARs to the set of already identified CARs for the specific municipality. Since CAR cancellations are tough to enforce and have been observed to always have at least one conflicting land claim with another CAR, we include cancelled CARs into our calculations.

These are the variables in the final irregular land-use data-set at the municipality-year level.

[15] Of the 559 municipalities in the Amazon biome, 557 municipalities have been fully processed. Within these processed municipalities, we observe 891,234 unique CARs in the raw data-files and successfully process 829,260 CARs. Additional work needs to be done to understand this slippage.

- car_union_area
- car_area_intersect_indi
- car_area_intersect_conserve
- car_area_intersect_forest_total
- car_area_intersect_forestA
- car_area_intersect_forestB
- car_area_intersect_forestC
- car_area_cancelled
- car_area_notcancelled
- year
- municipio

D Irregular land-use variables visualized

Figure **??** presents the area of the union of all CARs within a municipality for each year of our data. Here, we can see the evolution of both the amount of claimed land and the system itself, which only covers the whole Amazon biome starting in 2015. Figures **??** and **??** present the total areas of intersection between the areas in **??** and indigenous and conservation lands. Allcloropleths use the same keys, so comparison across figures is straightforward. We have not included the forested land figure nor the cancelled vs non-cancelled land figures because they are still begin finalized.

Appendix C Spatial Resolution Algorithm

Resolving Spatial Conflicts Between Rural Properties— Rural property lines in the CAR database are self-reported, and thus subject to errors and inconsistencies. As a consequence, many property claims overlap spatially. To avoid double-counting, we must therefore resolve these overlaps, a nontrivial task since it is in practice not possible to determine which CAR is the "correct" when there are conflicts.

We implement an algorithm

1. Drop properties with $< 10\%$ of deforested area in 2014
2. Set aside properties without any overlaps with other properties ($< 10\%$ of property area)
3. Apply the following spatial conflict resolution algorithm:
   - If i contains j (overlap $> 90\%$) and $> 80\%$ of its deforested area is in j, drop i
   - If i contains j and $> 80\%$ of its deforested area is in i, drop j
   - If i intersects with j (overlap $< 90\%$) and $> 80\%$ of its deforested area is in the intersection, but not for j, drop i
   - If i intersects with j and $> 80\%$ of their deforested areas are outside the intersection, keep i and j and randomly remove the intersection area from i or j
   - If i intersects with j and $> 80\%$ of their deforested areas are in the intersection, randomly keep i or j
4. Combine adjusted properties with those set aside in 2.

FIGURE B.1: Illustration of the Spatial Conflict Resolution Algorithm

output/spatial_rez_algo.pdf

NOTES— The figure illustrates

Figure B.2 illustrates step 3 above for all relevant rural properties a Brazilian municipality. The plot on the left shows property lines as entered in the CAR system. Multiple property lines cross with those from neighbors and there are several clusters including multiple seemingly independent properties contained in a larger property claim. After applying the resolution algorithm, the adjusted property boundaries look like the second plot. There is a clear decrease in the density of polygons, as many of those were spurious or would result in double-counting of deforestation.

Assigning Treatment Status— After implementing the spatial resolution approach described above, we assign treatment status to properties located in the eligible federal land parcels (amnesty target areas), as described in the main text of the paper. Specifically for the 2009 Amnesty, the treatment status of parcel i is defined by the rule $w = \mathbb{1}\{\text{occupied}_i \leq 2004\} \times \mathbb{1}\{\text{area}_i < 1{,}500\}$. We define a parcel as

FIGURE B.2: Illustration of the Spatial Conflict Resolution Algorithm

output/algo_example.pdf

NOTES— The figure illustrates

occupied in year t if at least 10% of its legacy forest has been deforested. This is a high-resolution, precisely measured way to assign occupation. Land ownership claims in Brazil principally follow the concept of *Uti possidetis* (ownership by use), where the most common strategy to establish "use" is to deforest portions of the land (signaling anthropomorphic action). Even fencing, which is costly and without proper monitoring can be easily cut down or altered, requires forest cleaning prior to installation.

Thus, the measurement error from $\mathbb{1}\{\text{occupied}_i \leq 2004\}$ is likely small — forest clearing without occupation could still enable the individual to claim occupation ex-post and occupation without forest cleaning would be unlikely and difficult to grant the individual eligibility status for the forgiveness policy.

The second eligibility criteria is more difficult to measure directly. In principle, land grabbers have an incentive to declare as-large-as-possible property boundaries in the CAR system, since they can use the registry itself as supplemental "evidence" in a case for ownership. However, entering a new property registry with boundaries in conflict with pre-existing properties could delay or invalidate the technical approval of registry. Because the CAR system and therefore property lines was introduced after the 2009 Forgiveness Law, declared property lines in the system were not binding claims at the time of the policy introduction. That is to say, if a property has 50,000 hectares in CAR, in theory the property claimant could portion off a section of area under the policy threshold.

To address the potential measurement error introduced from using property areas as declared, we establish a simple occupation rule: if the total deforested area in a property is above 1,500 hectares, that parcel is considered as if $\mathbb{1}\{\text{area}_i \geq 1,500\}$. The idea is that if deforestation signals occupation,

Appendix D Robustness and Additional Results: 2009 Land-Grabbing Amnesty

Overview

This section details the procedure used to construct the enforcement intensity estimators, as implemented in the provided `R` script. The process involves two primary stages. First, we establish a link between satellite-detected deforestation warnings and subsequently issued environmental fines using a spatio-temporal matching algorithm. Second, using these matched pairs, we compute both a naive enforcement rate and a robust rate that adjusts for satellite observation censorship due to cloud cover, as discussed in the paper.

Stage 1: Spatio-Temporal Matching of Fines to Warnings

To measure enforcement, we first identify which deforestation warnings plausibly triggered a corresponding environmental fine. A match is established if a fine is issued within a proximate spatial and temporal window following a warning.

Spatial Matching Criterion

The spatial link is created by generating a circular area of influence around each deforestation warning polygon. The provided `R` code implements this by first transforming the coordinate reference system to a metric projection (EPSG:3857) to allow for distance-based calculations.

For each warning, a circular buffer is created. The radius (r) of this buffer is determined by a pre-defined area (A), ensuring that each buffer represents a constant search area. The analysis is conducted for multiple buffer areas (e.g., 0.5, 1.0, and 2.0 hectares). The radius is calculated as:

$$r = \sqrt{\frac{A}{\pi}}$$

A fine, represented by its geographic point coordinates, is considered a spatial match if it falls within the boundary of a warning's circular buffer.

Temporal Matching Criterion

A spatially matched fine is only considered a valid enforcement action if its date of issuance occurs within a pre-defined time window following the warning's observation date. The script defines this window as being between the day of the warning and six months after:

$$t_{warning} \leqslant t_{fine} \leqslant t_{warning} + 6 \text{ months}$$

The total count of fines satisfying both criteria in a given region g (e.g., control or target area) and time period t is denoted as $F_{g,t}$.

Stage 2: Construction of Enforcement Estimators

The Naive Enforcement Estimator

The naive estimator of enforcement intensity, $\hat{\pi}_{g,t}$, is the simple ratio of the number of matched fines ($F_{g,t}$) to the total number of observed warnings ($W_{g,t}$) in a given region and time period.

$$\hat{\pi}_{g,t} = \frac{F_{g,t}}{W_{g,t}} \tag{8}$$

This estimator is biased when $W_{g,t}$ is censored by factors such as cloud cover, as it underestimates the true extent of detectable illegal activity.

The Cloud-Adjusted Enforcement Estimator

To correct for observational censorship, we construct a cloud-adjusted estimator, $\hat{\pi}^{cloud}_{g,t}$, following the logic outlined in the paper. This procedure relies on the assumption that the rate of illegal activity is uniform across both clear and cloud-covered areas within a region.

Step 1: Calculate Visible Area Fraction ($v_{g,t}$) For each region g and time period t, the program calculates the fraction of the total area that is observable (i.e., cloud-free). This is done by spatially intersecting the region's polygon with shapefiles of cloud-free areas. The visible fraction, $v_{g,t}$, is the ratio of the resulting clear area to the total regional area:

$$v_{g,t} = \frac{\text{Area}(g \cap \text{ClearSpot}_t)}{\text{Area}(g)}$$

Step 2: Compute Cloud-Adjusted Enforcement Rate ($\hat{\pi}^{cloud}_{g,t}$) Finally, the cloud-adjusted enforcement rate is computed by dividing the number of matched fines by the estimated total (not just observed) number of warnings.

$$\hat{\pi}^{cloud}_{g,t} = \frac{F_{g,t}}{\hat{W}^{total}_{g,t}} = \frac{F_{g,t}}{(W_{g,t}/v_{g,t})} \tag{9}$$